



ISHLT MEETING

Predicting expiration among potential donation after circulatory death donors at 30- and 120-min post extubation¹

John P. White, BA,^a Joseph Song, BS,^a Peter D. Cho, BS,^b Hedwig Zappacosta, BS,^a Stephanie McKay, MPH,^a Alexey Abramov, MD,^c Malini Daniel, MD,^c Tom Seto,^d Sharon West,^e Donatello Telesca, PhD,^f Majid Husain, MD,^{a,g} Angshuman Saha, PhD,^f and Abbas Ardehali, MD^{a,c}

From the ^aDavid Geffen School of Medicine, University of California Los Angeles, Los Angeles, California; ^bDrexel University College of Medicine, Philadelphia, Pennsylvania; ^cDivision of Cardiac Surgery, Department of Surgery, University of California Los Angeles, Los Angeles, California; ^dOneLegacy, Azusa, California; ^eGift of Life Donor Program, Philadelphia, Pennsylvania; ^fDepartment of Biostatistics, University of California Los Angeles, Los Angeles, California; and the ^gDivision of Pediatric Cardiology, Department of Pediatrics, UCLA Mattel Children's Hospital, Los Angeles, California.

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PURPOSE: Many potential donation after circulatory death (DCD) donors do not progress to circulatory death within meaningful timeframes, leading to wasted procurement resources. We therefore sought to develop robust machine learning models to predict donor expiration at 30- and 120-min following extubation.

METHODS: We performed a retrospective cohort study using data from 3 organ procurement organizations, analyzing 4464 potential DCD donors from 2014-2025 across 61 clinical variables. Primary endpoints were binary expiration at 30- and 120-min post extubation. Three cohorts were constructed. Regression analysis utilized the “complete-case dataset”, while “max sample” (n = 4464; 14 variables) and “max variable” (n = 2092; 61 variables) datasets were used for XGBoost modeling to maintain completeness > 85%. Results were evaluated using receiver operating characteristic (ROC) and precision-recall (PR) area under the curve (AUC).

RESULTS: Regression modeling yielded modest results. Among the machine learning models, the max variable model had the highest performance and yielded ROC-AUCs of 76% and 89% for 30- and 120-min, respectively, while PR-AUCs reached 80% and 97%. In both the 30- and 120-min models, final cough, lung offer, arterial blood gas pH, and serum platelets were among the top 6 variables of importance.

Abbreviations: AUC, Area under curve; DBD, Donation after brain death; DCD, Donation after circulatory death; OPO, Organ Procurement Organization; PR, Precision-recall; ROC, Receiver operating characteristics; UNOS, United Network for Organ Sharing; US, United States; XGBoost, Extreme Gradient Boosting

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corresponding author: Abbas Ardehali, MD, Surgery and Medicine, 62-186 CHS 10833 Le Conte Avenue, Los Angeles, CA 90095. Telephone: +310-825-5841. Fax: +310-825-7473.

E-mail address: ardehali@mednet.ucla.edu.

CONCLUSION: These models represent the most sophisticated approach to predicting DCD donor expiration to date. By leveraging a large multicenter cohort and nonlinear machine-learning techniques, our model more accurately captures the complex physiologic dynamics governing donor progression. While further validation is warranted, this framework has potential to reduce futile procurements and more accurately allocate resources, expanding access to life-saving transplantation. *J Heart Lung Transplant*

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Transplantation remains the gold standard therapy for end-stage heart and lung failure; however, persistent donor shortages limit access to life-saving therapy in the United States (US). In response, national efforts have focused on expanding the donor pool by accessing underutilized populations such as extended criteria donor organs.¹ Additionally, transplant centers have advocated for the expansion of donation after circulatory death (DCD) procurement to augment traditional donation after brain death (DBD) donation. This initiative has resulted in substantial year-over-year growth in DCD recovery, including more than 7200 DCD donors in 2025.²

Despite this progress, DCD utilization for thoracic transplantation remains disproportionately limited. The principal barrier is the unpredictability of the agonal phase following extubation. For heart and lung recovery, circulatory arrest must occur within relatively narrow time windows, ideally 30 min for cardiac viability³ and less than 120 min for pulmonary viability.⁴ Failure to meet these thresholds results in non-procurement, unnecessary mobilization of procurement teams, increased cost, and emotional burden to donor families. Although DCD donors comprised 37% of kidney transplants in 2023,⁵ they represented a much smaller fraction of heart and lung transplants.⁶

Prior efforts to predict timely expiration have relied on relatively small cohorts and limited physiologic variables, using regression-based approaches. The DCD-N score, which incorporated neurological reflexes and oxygenation parameters, demonstrated acceptable discrimination in early studies but showed limited calibration and generalizability in larger external cohorts.⁷ The University of Colorado model was another attempt using univariate analysis, but on DonorNet data from a single organ procurement organization (OPO). While they claimed excellent discrimination, the model was only trained on a sample of 140 donors and similarly lacked greater generalizability.⁸ Consequently, these models provide only modest predictive performance and fail to account for the complex, nonlinear physiologic interactions that govern progression to circulatory arrest following extubation.

More recently, investigators from Stanford and collaborating centers developed a machine-learning model using multicenter data to predict donor progression and reduce futile procurements.⁹ However, this model was developed specifically in the context of liver transplantation and may not be directly applicable to thoracic organ procurement, where acceptable warm ischemic thresholds and logistical considerations differ substantially. The researchers also

chose to use a chatbot to scrape patient records and develop the dataset which carries inherent risk of hallucination and misattributed data. In addition, the model generates a binary decision, which limits the ability of procurement teams to tailor risk tolerance based on clinical urgency, travel logistics, and institutional resources, factors that frequently influence real-world procurement decisions. As a result, many institutions continue to rely heavily on subjective clinical gestalt rather than reproducible, data-driven guidance when determining whether to pursue a potential DCD donor.

Given the clinical and economic stakes, there is a compelling need for robust predictive tools capable of accurately forecasting DCD donor expiration within clinically meaningful timeframes. Accordingly, our primary endpoint was to develop and validate a multi-institutional machine learning model to predict DCD donor expiration at 30- and 120-min post extubation.

Methods

Study design

This is a retrospective study using data collected from 3 geographically distinct OPOs (HonorBridge, Chapel Hill, NC; Gift of Life Donor Program, Philadelphia, PA; OneLegacy, Azusa, CA). The study period spanned January 2014 through June 2025. All potential DCD donors for whom withdrawal of life-sustaining therapy was initiated with intent for organ donation were eligible for inclusion. This study was deemed exempt from review by the Institutional Review Board at the University of California, Los Angeles.

A total of 4464 potential DCD donors were identified. A standardized set of clinically relevant variables was defined a priori, and each OPO extracted these variables directly from their internal databases. Data was provided in structured spreadsheets and subsequently consolidated. A total of 61 clinical variables were collected, including demographic characteristics, prior history, neurologic assessments, ventilatory parameters, laboratory values, and hemodynamic measurements at the time preceding organ offer. Variables were selected based on clinical plausibility and availability across participating OPOs. Neurologic variables are titled “initial” or “final” based on when they were chronologically collected and were completed during evaluation, well prior to extubation. Additional “differential” parameters are the difference between the last and first measurement to capture clinical improvement or

deterioration. All other variables are the last collected measurement to mimic the available data a clinician would have during procurement decisions. Since donor offer timing varies across organs and re-offer scenarios, a standardized datapoint prior to offer could not be defined. The primary endpoints were expiration at 30- and 120-min post extubation. Data for the secondary endpoints of expiration at 60- and 90-min post extubation are provided in the supplemental tables and figures.

Cohort construction

To preserve sample size, 3 cohorts were constructed. The complete-case dataset from all 3 OPOs was utilized for a multivariable logistic regression to develop a baseline model. For machine learning modeling, the max sample cohort included 4464 potential donors from all 3 OPOs and 14 variables that met $\geq 85\%$ completeness threshold for inclusion. However, we also wanted to use the breadth of variables to develop an alternate model from 2 OPOs which resulted in 61 variables but was restricted to 2092 donors to maintain completeness criteria. Each dataset was randomly partitioned into training (80%) and test (20%) sets.

Regression model development

Univariate logistic regression primarily served as an exploratory tool to study the predictive power of individual variables. As a result, each univariate model was based on different sample sizes to encompass the entire dataset and did not require imputation. The subsequent multivariate logistic model was fit on variables selected through a stepwise Akaike Information Criterion model selection procedure during which median and mode values were used for imputation of continuous and categorical variables, respectively.

Machine learning model development

To ensure robust statistical analysis, variables were required to meet a $\geq 85\%$ completeness threshold for inclusion in machine learning modeling. Missing values in numeric variables were imputed using medians and those in the categorical variables were imputed using the mode. Supervised machine learning models were independently developed for each endpoint within both cohorts. Extreme Gradient Boosting (XGBoost) algorithm was selected due to its capacity to model nonlinear relationships and complex variable interactions. Variables were ranked according to importance and variables with $< 5\%$ impact of the highest ranked variable were discarded and the model was re-run to analyze performance metrics. This minimizes the number of extraneous variables with minor impact to the final model, reducing the burden of excess variable collection.

Model evaluation

Model performance for regression and machine learning models was assessed on the held-out test sets.

Discriminatory ability was evaluated using area under the curve (AUC) for the receiver operating characteristics (ROC) and precision-recall (PR) curves. Feature importance analyses were also performed to assess model stability and interpretability. Lastly, calibration curves and Brier scores are provided in [Supplemental Figures 3 and 4](#).

Web application

We developed an interactive web-based application using R Shiny to operationalize the DCD prediction model, enabling real-time input of clinical variables and generation of predicted probabilities. The application is available at (<https://rsc-domstat.med.ucla.edu/dcdscore/>).

Results

Variables of interest among potential DCD donors

A total of 4464 potential DCD donors were included in the complete-case and max sample datasets. Baseline characteristics of this population and expiration outcomes are presented in [Table 1](#). A total of 2713 (61%) and 3588 (80%) donors expired within 30 and 120 min respectively. The median donor age was 46 years (34-56), and 37% of donors were female. Comorbid conditions included a history of hypertension in 40% of donors and diabetes in 18%. The max variable cohort was titrated to include 2092 donors with high data completeness, incorporating a larger set of clinical variables. Donor characteristics were largely similar to the max sample cohort, with a median age of 47 years (35-56) and 39% female donors.

Univariate analysis predicting expiration among potential DCD donors

Univariate logistic regression analyses identified variables associated with donor expiration within 30 min following extubation ([Table 2](#)). The presence of preserved neurologic reflexes, including cough, gag, corneal, pupillary, and pain responses, as well as spontaneous respirations, were consistently associated with a lower likelihood of expiration within 30 min (all $p < 0.001$). In contrast, physiologic and laboratory parameters reflecting greater physiologic instability were associated with increased likelihood of early expiration, including lower arterial blood gas pH, systolic and diastolic blood pressures, mean arterial pressure, higher heart rate, and serum sodium levels (all $p < 0.001$).

A similar pattern of associations was observed for expiration within 120 min following extubation ([Table 3](#)). Preserved neurologic reflexes and spontaneous respirations remained strongly associated with a lower likelihood of expiration within this extended time window (all $p < 0.001$). Conversely, markers of physiologic compromise, including lower blood pressures, arterial blood gas pH, higher heart rate, ventilator requirements, and serum sodium, were associated with increased likelihood of expiration within

Table 1 Baseline Characteristics For All Datasets

Variable	Complete-Case N = 4464	Max Sample N = 4464	Max Variable N = 2092
<i>Patient Demographics</i>			
Age, Year	46 (34-56)	46 (34-56)	47 (35-56)
Gender, Female	1632 (37%)	1632 (37%)	806 (39%)
Race, White	2727 (77%)	N/A	1645 (79%)
Weight, kg	86 (72-103)	N/A	84 (69-101)
History of Diabetes, Yes	829 (18%)	829 (18%)	343 (16%)
History of Hypertension, Yes	1845 (40%)	1845 (40%)	860 (41%)
Are They Insulin Dependent? Yes	244 (5%)	N/A	154 (7%)
Cause of Death, Anoxia	2216 (50%)	N/A	1365 (65%)
Mechanism of Death, Cardiovascular	1353 (30%)	N/A	828 (40%)
Circumstance of Death, Natural Causes	1999 (45%)	N/A	1182 (57%)
Was a Lung Offer Made? Yes	1184 (27%)	1184 (27%)	788 (38%)
<i>Neurological Results</i>			
Initial Pupillary Reaction, Present	1720 (39%)	N/A	1132 (56%)
Initial Cough Reaction, Present	1532 (34%)	N/A	977 (48%)
Initial Gag Reaction, Present	1110 (25%)	N/A	738 (37%)
Initial Corneal Reaction, Present	1032 (23%)	N/A	693 (34%)
Initial Pain Response Reaction, Present	1362 (31%)	N/A	918 (45%)
Initial Spontaneous Respirations, Present	1876 (42%)	N/A	1264 (63%)
Final Pupillary Reaction, Present	1961 (44%)	1961 (44%)	1164 (56%)
Final Cough Reaction, Present	1936 (43%)	1936 (43%)	1133 (55%)
Final Gag Reaction, Present	1433 (32%)	1433 (32%)	802 (39%)
Final Corneal Reaction, Present	1358 (30%)	1358 (30%)	811 (39%)
Final Pain Response Reaction, Present	1659 (37%)	1659 (37%)	1011 (47%)
Final Spontaneous Respirations	2301 (52%)	N/A	1553 (75%)
Change in Neurological Findings	0 (−1 to 2)	N/A	0 (0-1)
<i>Hemodynamic Results</i>			
Systolic Blood Pressure, mmHg	121 (105-137)	N/A	121 (105-138)
Diastolic Blood Pressure, mmHg	65 (55-76)	N/A	66 (56-75)
Heart Rate, BPM	90 (77-105)	N/A	91 (77-109)
Mean Arterial Pressure, mmHg	83 (71-94)	N/A	84 (73-95)
<i>Urinalysis Results</i>			
Urinalysis Color, Yellow	2413 (54%)	N/A	1451 (69%)
Urinalysis Appearance, Cloudy	817 (18%)	N/A	480 (23%)
Urinalysis pH	6.0 (4.2-7.2)	N/A	5.5 (5.0-6.0)
Urinalysis Specific Gravity	1.02 (1.02-1.03)	N/A	1.02 (1.02-1.03)
Urinalysis Red Blood Cells, Positive	2483 (56%)	N/A	1521 (73%)
Urinalysis White Blood Cells, Positive	2447 (55%)	N/A	1536 (74%)
Urinalysis Epithelial Cells, Positive	1269 (28%)	N/A	768 (37%)
Urinalysis Casts, Positive	906 (20%)	N/A	582 (29%)
Urinalysis Glucose, Positive	782 (18%)	N/A	508 (24%)
Urinalysis Bacteria, Positive	1564 (35%)	N/A	934 (45%)
<i>Serum Results</i>			
Blood Group, O	1382 (31%)	N/A	983 (47%)
Serum Arterial O ₂ , %	98 (95-100)	N/A	98 (96-100)
Serum Hemoglobin	9.9 (8.0-11.6)	N/A	9.8 (8.3-11.5)
Serum Sodium	144 (139-150)	144 (139-150)	144 (139-149)
Serum White Blood Cell Count	12.6 (9.3-17.2)	N/A	12.6 (9.2-17.3)
Serum Hematocrit	30 (26-36)	N/A	30 (25-35)
Serum Platelets	189 (123-271)	189 (123-271)	187 (121-273)
Serum Red Blood Cell	3.3 (2.9-3.9)	N/A	3.3 (2.8-3.9)
Arterial Blood Gas pH	7.42 (7.35-7.46)	7.42 (7.35-7.46)	7.42 (7.34-7.47)
Arterial Blood Gas pO ₂ mm Hg	116 (83-227)	116 (83-227)	123 (84-245)
Arterial Blood Gas Bicarbonate	25 (22-28)	N/A	24 (21-27)
Arterial Blood Gas Base Excess	0.7 (−3.0 to 3.7)	N/A	0.4 (−3.2 to 3.4)
<i>Ventilator Results</i>			
FiO ₂ , %	50 (40-80)	N/A	50 (40-80)
Peep, cmH ₂ O	7 (4-10)	N/A	8 (5-10)

(continued on next page)

Table 1 (Continued)

Variable	Complete-Case N = 4464	Max Sample N = 4464	Max Variable N = 2092
<i>Differential Results</i>			
Serum White Blood Cell Differential	-0.3 (-4.8 to 4.3)	N/A	-0.1 (-5.3 to 4.4)
Serum Hemoglobin Differential	-2.6 (-4.2 to (-1.3)	N/A	-2.6 (-4.3 to (-1.0))
Serum Hematocrit Differential	-8.6 (-13.0 to (-3.4))	N/A	-8.5 (-13.4 to (-3.6))
Serum Platelet Differential	-42 (-110 to 18)	N/A	-41 (-110 to 18)
Serum Red Blood Cell Differential	-0.9 (-1.7 to (-0.1))	N/A	-0.9 (-1.4 to (-0.3))
Serum Sodium Differential	5 (0-10)	N/A	5 (0-11)
Mean Arterial Pressure Differential	0.0 (-5.1 to 21.0)	N/A	0.7 (-5.3 to 20.7)
Arterial Blood Gas pH Differential	0.1 (0.0-0.2)	N/A	0.1 (0.0-0.2)
Expired at 30 min	2713 (61%)	2713 (61%)	1243 (59%)
Expired at 120 min	3588 (80%)	3588 (80%)	1681 (82%)

Values are expressed as frequency (proportion of population) or mean (interquartile range).

See supplemental Table 1 for variable data missingness

120 min. Additional laboratory markers, including lower platelet counts and arterial oxygenation demonstrated significant associations with expiration. Younger age continued to be associated with increased likelihood of expiration.

Multivariable model predicting expiration among potential DCD donors

Multivariable regression analysis identified independent predictors of expiration following extubation with results displayed in Table 4. In the 30-min model, neurologic findings, including final pupillary reaction, final cough reaction, and final spontaneous respirations were associated with a decreased likelihood of expiration. Physiologic and laboratory variables associated with increased likelihood of expiration included weight and heart rate, while lower urinalysis pH, serum sodium differential, and systolic blood pressure were also independently associated with expiration within 30 min. Additional predictors included serum platelet differential, serum platelets, urinalysis glucose, and circumstance of death categorized as other.

A similar set of variables remained significant in the multivariable model predicting expiration within 120 min. Independent predictors included change in neurological findings, serum white blood cell differential, serum arterial blood gas pH differential, systolic blood pressure, final pupillary reaction, serum platelet differential, urinalysis pH, arterial blood gas pH, final spontaneous respirations, initial pain response reaction, weight, serum platelets, serum sodium, arterial blood gas pO₂, initial gag reaction, circumstance of death (non-motor vehicle accident) and history of diabetes greater than 10 years.

XGBoost model predicting expiration among potential DCD donors

Model discrimination for prediction of donor expiration at 30 and 120 min following extubation is summarized in Table 5, with ROC and PR curves shown in Figures 1 and 2. For prediction of expiration within 30 min, the multivariable logistic

regression model demonstrated moderate discrimination with an ROC-AUC of 0.77 and PR-AUC of 0.81. The XGBoost max sample model achieved ROC-AUC values of 0.74 in the training cohort and 0.72 in the testing cohort, with PR-AUC values of 0.80 and 0.78, respectively. Incorporation of additional variables in the max variable model improved discrimination, yielding ROC-AUC values of 0.80 in training and 0.76 in testing, with PR-AUC values of 0.83 and 0.80, respectively.

Variable importance analysis identified several key predictors of early expiration and is displayed in Table 6. In the 30-min max sample model, the most influential variables included arterial blood gas pH (gain 0.507), lung offer status (0.199), sodium (0.183), arterial blood gas pO₂ (0.086), and age (0.024). In the 30-min max variable model, important predictors included final cough reaction (0.238), arterial blood gas pH (0.380), serum platelets (0.156), lung offer status (0.156), and final pain response reaction (0.128).

Model performance was higher for prediction of expiration within 120 min. The multivariable regression model demonstrated an ROC-AUC of 0.81 and PR-AUC of 0.94. The XGBoost max sample model achieved ROC-AUC values of 0.84 in training and 0.83 in testing, with PR-AUC values of 0.96 and 0.95. The XGBoost max variable model demonstrated the highest discrimination with ROC-AUC values of 0.90 in training and 0.89 in testing, and PR-AUC values of 0.97 and 0.97.

Variable importance analysis for the 120-min models demonstrated similar predictors. In the max sample model, the most influential variables included lung offer status (gain 0.383), arterial blood gas pH (0.380), serum platelet count (0.147), serum sodium (0.062), and arterial blood gas pO₂ (0.029). In the max variable model, lung offer status (0.447), arterial blood gas pH (0.309), serum platelet count (0.123), final gag reaction (0.036), and final pain response reaction (0.035) were the strongest contributors (Table 5).

Discussion

In this retrospective study of 4464 potential DCD donors, we developed and validated the most sophisticated

Table 2 Univariate Regression Analysis From The Complete-Case Dataset at 30 min Post Extubation

Variable	p-Value	Dependency
Final Cough Reaction	< 0.001	Present -> No Expiration
Final Gag Reaction	< 0.001	Present -> No Expiration
Final Corneal Reaction	< 0.001	Present -> No Expiration
Final Pupillary Reaction	< 0.001	Present -> No Expiration
Final Pain Response Reaction	< 0.001	Present -> No Expiration
Serum Platelets	< 0.001	Lower Value -> Yes Expiration
Arterial Blood Gas pH	< 0.001	Lower Value -> Yes Expiration
Final Spontaneous Respirations	< 0.001	Present -> No Expiration
Serum Platelet Differential	< 0.001	Lower Value -> Yes Expiration
Was a Lung Offer Made?	< 0.001	Yes Value -> Yes Expiration
Initial Cough Reaction	< 0.001	Present -> No Expiration
Serum Sodium	< 0.001	Higher Value -> Yes Expiration
FiO ₂	< 0.001	Higher Value -> Yes Expiration
Initial Gag Reaction	< 0.001	Present -> No Expiration
Initial Corneal Reaction	< 0.001	Present -> No Expiration
Initial Pain Response Reaction	< 0.001	Present -> No Expiration
Initial Spontaneous Respirations	< 0.001	Present -> No Expiration
Change in Neurological Findings	< 0.001	Lower Value -> Yes Expiration
Arterial Blood Gas Base Excess	< 0.001	Lower Value -> Yes Expiration
Initial Pupillary Reaction	< 0.001	Present -> No Expiration
Heart Rate	< 0.001	Higher Value -> Yes Expiration
Peep	< 0.001	Higher Value -> Yes Expiration
Systolic Blood Pressure	< 0.001	Lower Value -> Yes Expiration
Weight	< 0.001	Higher Value -> Yes Expiration
Serum Arterial O ₂	< 0.001	Lower Value -> Yes Expiration
Urinalysis Glucose	< 0.001	Present -> Yes Expiration
Arterial Blood Gas Bicarbonate	< 0.001	Lower Value -> Yes Expiration
Urinalysis Appearance	< 0.001	Cloudy, Other Value -> Yes Expiration
Mean Arterial Pressure	< 0.001	Lower Value -> Yes Expiration
Diastolic Blood Pressure	< 0.001	Lower Value -> Yes Expiration
Ventilator Mode	< 0.001	PRVC, SIM Value -> Yes Expiration
Urinalysis pH	< 0.001	Lower Value -> Yes Expiration
Arterial Blood Gas pO ₂	< 0.001	Lower Value -> Yes Expiration
Serum Hematocrit	< 0.001	Higher Value -> Yes Expiration
Age	< 0.001	Lower Value -> Yes Expiration
Urinalysis Casts	< 0.001	Present -> Yes Expiration
Serum Hemoglobin	< 0.001	Higher Value -> Yes Expiration
Gender	0.001	Male Value -> Yes Expiration
Serum Red Blood Cell Count	0.002	Higher Value -> Yes Expiration
History of Hypertension	0.003	Yes at 0-5, 6-10 Years -> Yes Expiration
Mechanism of Death	0.008	Blunt Injury, Other Value -> Yes Expiration
Serum White Blood Cell Count	0.009	Higher Value -> Yes Expiration
Serum Hematocrit Differential	0.02	Higher Value -> Yes Expiration
Circumstance of Death	0.02	Non-MV Accident, MVA -> Yes Expiration
Serum Sodium Differential	0.03	Higher Value -> Yes Expiration
Urinalysis Bacteria	0.04	Present -> Yes Expiration

See supplemental [Table 1](#) for variable data missingness

approach to predicting DCD donor expiration to date. Our max variable model utilized machine learning techniques, resulting in strong predictive performance with higher AUC than any extant model. These models were trained with the largest and most comprehensive known DCD database from 3 geographically distinct OPOs which was vital in overcoming the limitations of prior efforts and may enhance the generalizability of the model. Lastly, our web application enables physicians to titrate non-procurement risk by adjusting the precision threshold, allowing decision thresholds

to be tailored to clinical urgency and procurement cost which is a capability not incorporated in prior models.

When compared with existing prediction tools, our model demonstrates superior predictive performance while addressing several key methodological limitations of prior approaches. Earlier tools, including the DCD-N score and the University of Colorado model, were derived using traditional linear regression methods applied to relatively small datasets with limited physiologic variables. While informative, these linear approaches assume fixed

Table 3 Univariate Regression Analysis From The Complete-Case Dataset at 120 min Post Extubation

Variable	p-Value	Dependency
Final Cough Reaction	< 0.001	Present -> No Expiration
Final Gag Reaction	< 0.001	Present -> No Expiration
Final Pain Response Reaction	< 0.001	Present -> No Expiration
Final Corneal Reaction	< 0.001	Present -> No Expiration
Final Pupillary Reaction	< 0.001	Present -> No Expiration
Arterial Blood Gas pH	< 0.001	Lower Value -> Yes Expiration
Serum Platelets	< 0.001	Lower Value -> Yes Expiration
Was a Lung Offer Made?	< 0.001	Yes Value -> Yes Expiration
Final Spontaneous Respirations	< 0.001	Present -> No Expiration
Serum Platelet Differential	< 0.001	Lower Value -> Yes Expiration
Serum Sodium	< 0.001	Higher Value -> Yes Expiration
FiO ₂	< 0.001	Higher Value -> Yes Expiration
Initial Pain Response Reaction	< 0.001	Present -> No Expiration
Initial Cough Reaction	< 0.001	Present -> No Expiration
Change in Neurological Findings	< 0.001	Present -> Yes Expiration
Initial Gag Reaction	< 0.001	Present -> No Expiration
Peep	< 0.001	Higher Value -> Yes Expiration
Systolic Blood Pressure	< 0.001	Lower Value -> Yes Expiration
Mean Arterial Pressure	< 0.001	Lower Value -> Yes Expiration
Initial Spontaneous Respirations	< 0.001	Present -> No Expiration
Heart Rate	< 0.001	Higher Value -> Yes Expiration
Initial Corneal Reaction	< 0.001	Present -> No Expiration
Diastolic Blood Pressure	< 0.001	Lower Value -> Yes Expiration
Initial Pupillary Reaction	< 0.001	Present -> No Expiration
Arterial Blood Gas Base Excess	< 0.001	Lower Value -> Yes Expiration
Urinalysis Glucose	< 0.001	Present -> Yes Expiration
Serum Arterial O ₂	< 0.001	Lower Value -> Yes Expiration
Urinalysis pH	< 0.001	Lower Value -> Yes Expiration
Ventilator Mode	< 0.001	NC, PRVC, SIMV Value -> Yes Expiration
Arterial Blood Gas pO ₂	< 0.001	Lower Value -> Yes Expiration
Weight	< 0.001	Higher Value -> Yes Expiration
Arterial Blood Gas Bicarbonate	< 0.001	Lower Value -> Yes Expiration
Urinalysis Appearance	< 0.001	Cloudy, Other -> Yes Expiration
Mechanism of Death	< 0.001	Blunt Injury, Other Value -> Yes Expiration
Age	< 0.001	Lower Value -> Yes Expiration
Blood Group	0.001	A1 Value -> Yes Expiration
Serum Sodium Differential	0.01	Higher Value -> Yes Expiration
Urinalysis Casts	0.01	Present -> Yes Expiration
Serum White Blood Cell Differential	0.01	Higher Value -> Yes Expiration
Serum Hematocrit	0.02	Higher Value -> Yes Expiration
Serum Hemoglobin	0.04	Higher Value -> Yes Expiration
Circumstance of Death	0.04	MVA Value -> Yes Expiration

See supplemental Table 1 for variable data missingness

relationships between predictors and outcomes and are poorly suited to capture the complex, nonlinear physiologic interactions that govern progression to circulatory arrest following extubation. In contrast, we applied an XGBoost machine-learning framework capable of modeling nonlinear relationships and higher-order interactions. As a result, our models demonstrate improved discriminatory performance across all endpoints and provide a more robust framework for predicting donor progression following extubation.

Our model was developed on the backbone of the largest, most diverse and most comprehensive known DCD dataset. The breadth of 61 variables allowed incorporation of a larger physiologic feature set than prior efforts, resulting in exploration of previously unstudied interactions. Additionally,

discrete data fields directly imported from OPO records minimizes ambiguity associated with free-text interpretation and improves data fidelity. Lastly, partnering with 3 geographically distinct OPOs allowed for a truly representative sample that more closely reflect the characteristics of the contemporary national DCD donor population. Consequently, models derived from this dataset are more likely to maintain predictive performance when applied across institutions and procurement environments with varying clinical practices.

Finally, an important advantage of the present model is its ability to support titratable clinical decision-making. Our web application displays 3 decision thresholds, high sensitivity, balanced, and high precision, each corresponding to increasing precision cutoffs and reflecting a different point

Table 4 Multivariable Regression From The Complete-Case Dataset at 30- and 120-Min Post Extubation

Variable	p-Value	Odds Ratio (95% CI)
<i>30 min</i>		
Weight	< 0.001	1.01 (1.00, 1.01)
Serum Platelet Differential	< 0.001	1.00 (1.00, 1.00)
Heart Rate	< 0.001	1.01 (1.00, 1.02)
Final Pupillary Reaction	< 0.001	0.54 (0.42, 0.89)
Final Cough Reaction	< 0.001	0.57 (0.45, 0.80)
Final Spontaneous Respirations	< 0.001	0.60 (0.99, 1)
Serum Platelets	0.001	1.00 (1.00, 1.00)
Urinalysis pH	0.005	0.84 (0.74, 0.95)
Serum Sodium Differential	0.01	0.99 (0.97, 1.00)
Systolic Blood Pressure	0.01	0.99 (0.99, 1.00)
Circumstance of Death: Other	0.01	0.62 (0.42, 0.68)
Urinalysis Glucose	0.02	1.37 (1.01, 1.77)
<i>120 min</i>		
Change in Neurological Findings	< 0.001	0.85 (0.78, 0.94)
Serum White Blood Cell Differential	< 0.001	1.03 (1.01, 1.06)
Serum Arterial Blood Gas pH Differential	< 0.001	4.63 (2.02, 10.61)
Systolic Blood Pressure	< 0.001	0.99 (0.98, 0.99)
Final Pupillary Reaction	< 0.001	0.52 (0.37, 0.73)
Serum Platelet Differential	0.001	1.00 (0.99, 1.00)
Urinalysis pH	0.002	0.79 (0.69, 0.92)
Arterial Blood Gas pH	0.003	0.16 (0.05, 0.53)
Final Spontaneous Respirations	0.004	0.47 (0.28, 0.79)
Initial Pain Response Reaction	0.005	0.64 (0.47, 0.87)
Weight	0.005	1.01 (1.00, 1.01)
Serum Platelets	0.006	1.00 (0.99, 1.00)
Serum Sodium	0.006	1.03 (1.01, 1.05)
Arterial Blood Gas pO ₂	0.008	1.00 (0.99, 1.00)
Initial Gag Reaction	0.008	0.65 (0.48, 0.90)
Circumstance of Death: Non-MV Accident	0.04	0.71 (0.52, 0.98)
History of Diabetes: Yes, > 10 years	0.04	0.59 (0.35, 0.98)

See supplemental Table 1 for variable data missingness

on the precision-recall curve. This allows clinicians to adjust risk tolerance according to clinical context. For instance, when evaluating a critically ill recipient, a physician may elect to accept a higher-risk procurement strategy to maximize the probability of organ procurement. In these situations, the model operates at a high sensitivity threshold so that potential donors likely to expire are rarely missed, even if this increases the number of unsuccessful procurements (low precision). Conversely, circumstances involving substantial travel costs, limited surgical resources, or marginal donor characteristics may favor a more conservative

approach. In these settings, the model can be run at a high precision threshold, where a positive prediction is generated only when donor expiration is very likely. Although this configuration produces fewer positive predictions, a positive prediction means the model is very certain the donor will expire. By enabling this flexible adjustment of risk tolerance, the model provides a practical decision-support tool that better reflects the nuanced logistical and clinical considerations that influence DCD procurements.

Several limitations warrant acknowledgment. The retrospective design introduces potential bias related to

Table 5 Model AUC Performance at 30- and 120-Min Post Extubation

Variable	Complete-Case	Max Sample	Max Variable
30 Min Model (ROC-Train)	0.77	0.74	0.80
30 Min Model (ROC-Test)		0.72	0.76
30 Min Model (PR-Train)	0.81	0.80	0.83
30 Min Model (PR-Test)		0.78	0.80
120 Min Model (ROC-Train)	0.81	0.84	0.90
120 Min Model (ROC-Test)		0.83	0.89
120 Min Model (PR-Train)	0.94	0.96	0.97
120 Min Model (PR-Test)		0.95	0.97

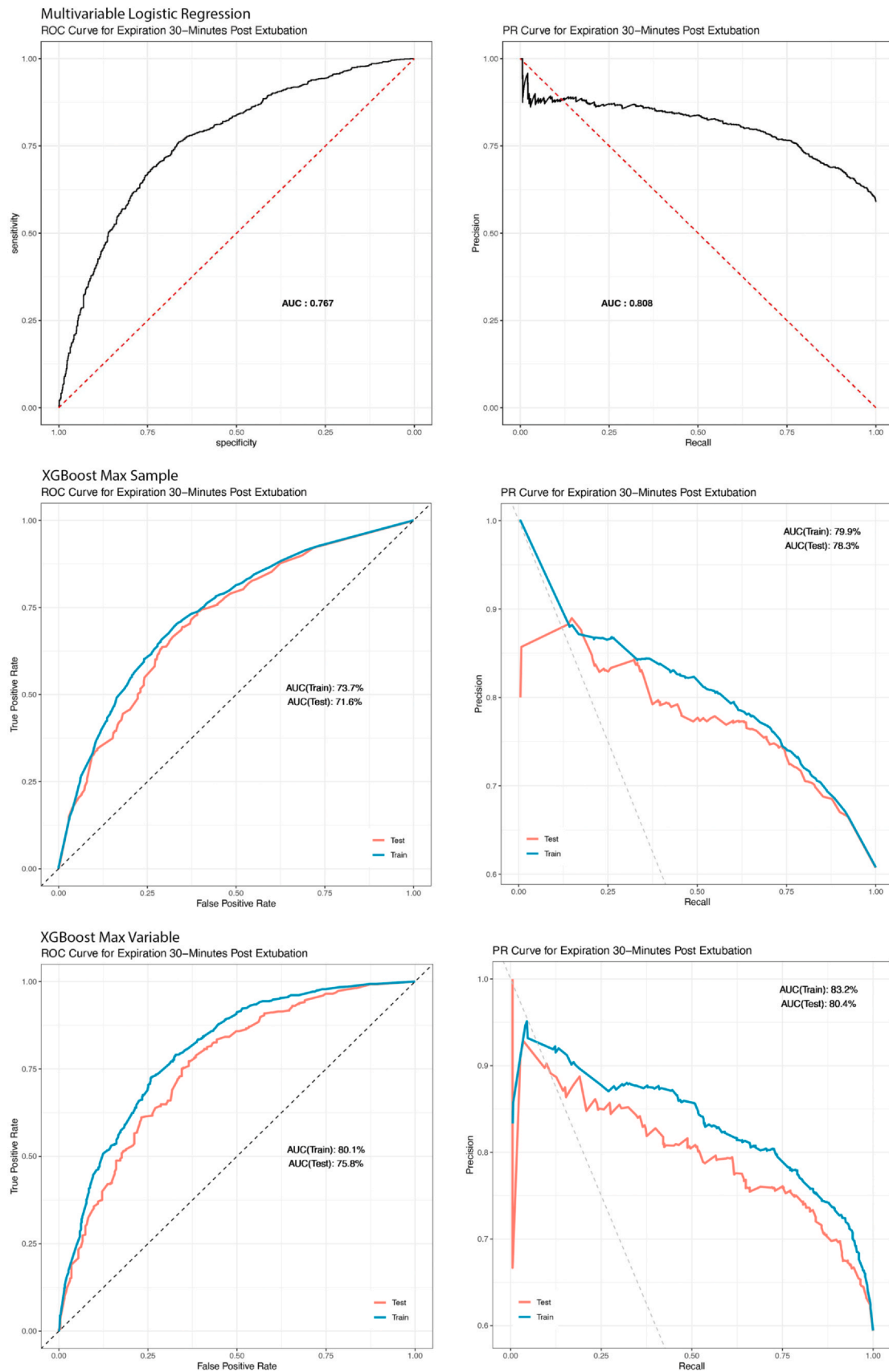


Figure 1 ROC and PR curves for multivariable regression, XGBoost max sample and XGBoost max variable models at 30 min post extubation. PR, precision-recall; ROC, receiver operating characteristic; XGBoost, Extreme Gradient Boosting

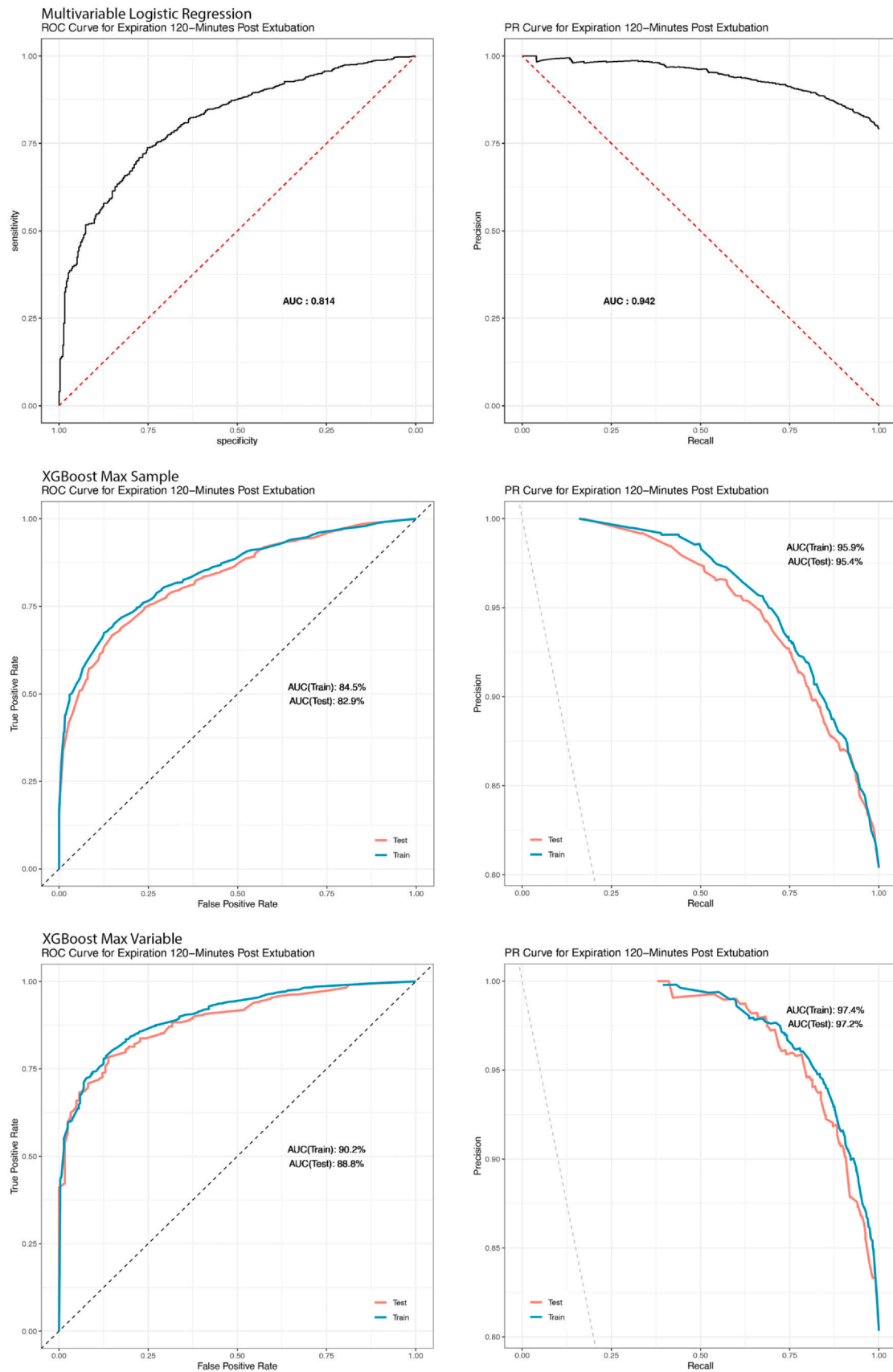


Figure 2 ROC and PR curves for multivariable regression, XGBoost max sample and XGBoost max variable models at 120 min post extubation. PR, precision-recall; ROC, receiver operating characteristic; XGBoost, Extreme Gradient Boosting

Table 6 XGBoost Variables of Importance From The Max Sample and Max Variable Models at 30- and 120-Min Post Extubation

Variable	Gain	Cover	Frequency
<i>30 Min Model (Max Sample)</i>			
#1 Arterial Blood Gas pH	0.507	0.425	0.324
#2 Was a Lung Offer Made?	0.199	0.161	0.118
#3 Serum Sodium	0.183	0.189	0.206
#4 Arterial Blood Gas pO ₂	0.086	0.171	0.265
#5 Age	0.024	0.055	0.088
<i>120 Min Model (Max Sample)</i>			
#1 Was a Lung Offer Made?	0.383	0.338	0.167
#2 Arterial Blood Gas pH	0.380	0.313	0.333
#3 Serum Platelets	0.147	0.186	0.300
#4 Serum Sodium	0.062	0.077	0.067
#5 Arterial Blood Gas pO ₂	0.029	0.086	0.133
<i>30 Min Model (Max Variable)</i>			
#1 Final Cough Reaction	0.238	0.170	0.088
#2 Arterial Blood Gas pH	0.160	0.172	0.176
#3 Serum Platelets	0.156	0.180	0.235
#4 Was a Lung Offer Made?	0.156	0.135	0.118
#5 Final Pain Response Reaction	0.128	0.134	0.088
#6 Serum Arterial O ₂	0.085	0.087	0.118
#7 Final Pupillary Reaction	0.049	0.061	0.059
#8 Systolic Blood Pressure	0.028	0.061	0.118
<i>120 Min Model (Max Variable)</i>			
#1 Was a Lung Offer Made?	0.447	0.334	0.152
#2 Arterial Blood Gas pH	0.309	0.298	0.303
#3 Serum Platelets	0.123	0.205	0.333
#4 Final Gag Reaction	0.036	0.067	0.061
#5 Final Pain Response Reaction	0.035	0.034	0.061
#6 Final Cough Reaction	0.029	0.024	0.030
#7 Systolic Blood Pressure	0.021	0.037	0.061

documentation and practice variability across OPOs. Although internally validated, the model has not yet undergone prospective testing, which will be essential to determine real-time clinical utility and impact on procurement efficiency. The model only incorporates variables that were provided by the OPOs. There may be relevant variables like ICU medications, past medical history, or pharmacological management at the time of extubation that could not be captured. Also, there may be impacts to variables, such as sedative administration effects on neurological variables, that we cannot control for. Additionally, the inputs used to train the model may not reflect the exact values at the time of procurement offer. Although cohorts were constructed to minimize missing data, residual missingness may still influence model performance and represents a practical limitation, particularly in clinical settings where all variables may not be readily available. Finally, while the model predicts likelihood of expiration within specified intervals, it does not assess post-transplant outcomes or graft quality.

Importantly, the present model should not be viewed as a static, one-time development, but rather as a dynamic framework that is expected to evolve alongside the rapidly expanding landscape of DCD donation. As the number of DCD donors continues to increase and the breadth and

granularity of data collected improves, so will the model. This will enable more robust training, improved feature representation, and enhanced model calibration. In parallel, ongoing refinement through hyperparameter optimization, incorporation of additional clinically relevant variables, and prospective validation will further strengthen predictive performance. Taken together, these iterative improvements position this model as a continually advancing decision-support tool, with the potential to achieve progressively greater accuracy and clinical utility over time.

In conclusion, we present the most comprehensive and highest-performing predictive model for DCD donor expiration developed to date. By leveraging a large multi-center cohort and nonlinear machine-learning techniques, our model more accurately captures the complex physiologic dynamics governing donor progression following extubation. While further validation is warranted, this framework has the potential to improve donor selection, reduce futile procurements, and ultimately expand access to life-saving heart and lung transplantation.

Conflicts of Interest

The authors declare no conflicts of interest.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.healun.2026.04.006](https://doi.org/10.1016/j.healun.2026.04.006).

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